

Audit of the Constraint Projection Framework

Four Incompatible Cutoffs, an Unsatisfiable Axiom, and a Curvature Functional Blind to the Functional Equation

Flag Condensate Collaboration
Sovereign-Stack ACS Research Program

August 29, 2026

Abstract

The Constraint Projection Framework (CPF) manuscript, archived as submitted at `papers/notes/Constraint_Pro`, claims a zero-free-parameter topological derivation of α , g , s , the dark-matter density, cosmological flatness and the Hubble radius from a single non-orientable surface. This note evaluates every load-bearing step. **Result: one displayed number survives its own computation**, and it is CHSH. The rest fails on four independent grounds: an algebraic error that a note already in this repository had corrected (§2); a scale parameter asserted to be fixed but required to take four mutually exclusive values spanning 117 decades (§3); an axiom with no model, because the surface its own clauses force has a finite mapping class group (§4); and two claims falsified in this repository’s Elimination Ledger on 2026-07-26 and restated here without reference (§5). Sections 6–9 treat the curvature functional, the Wigner’s-friend sum, the dark-matter identification, and Axioms I–II. §10 states what survives and what it would take to repair each leg. Instrument: `code/constraint_projection/cpf_audit.py`; artifact `docs/constraint_projection_audit.json`.

1 Scope and claim discipline

In scope: the algebra, topology and arithmetic of the manuscript’s §§2–8, evaluated against its own stated inputs; consistency with results already logged in this repository; the two citations that could be checked.

Out of scope: the interpretive material of §9 (relational measurement), which makes no falsifiable claim and is not assessed here; the philosophical framing of §1; questions of priority or authorship.

Standard applied. Every check below is written so that the manuscript could have passed it. No verdict is assumed by the instrument. Where a claim is *correct*, this note says so (§7 is the case).

2 §3: the fine-structure constant — a dropped factor of two

The manuscript’s §3 supplies three inputs and one conclusion:

$$C_{\mathcal{M}} = \frac{2\pi\epsilon_0 R}{L}, \quad L \equiv \ln(8R/a) + 1, \quad (1)$$

$$E_{\text{self}} = \frac{(e/2)^2}{2C_{\mathcal{M}}} = m_e c^2, \quad (2)$$

$$R = \frac{\hbar}{2m_e c}, \quad (3)$$

concluding $\alpha^{-1} = L$.

Proposition 1 (The conclusion does not follow). *Equations (1)–(3) imply $L\alpha = 2$, i.e.*

$$\alpha^{-1} = \frac{1}{2}(\ln(8R/a) + 1).$$

Claim	Tier	Mechanism of failure
$\alpha^{-1} = \ln(8R/a) + 1$	T4	factor of 2 dropped from the manuscript’s own self-energy match
zero free parameters	T4	a/R fitted per section; four incompatible values, 117 decades apart
Axiom III (ϕ_* parabolic)	T4	$\mathcal{M}$ is forced to be the Klein bottle; $\text{MCG}(K) = \mathbb{Z}/2 \oplus \mathbb{Z}/2$ is finite
$g = Sl = 2$, $s = Sl/4$	T4	already killed 2026-07-26; Sl carries one parity bit, and $g = 1$ for every closed curve
$k = 0 \Leftrightarrow \text{RH} \Leftrightarrow \Omega_k = 0$	T4	functional is invariant under $\beta \mapsto 1 - \beta$; on-line zeros give $k \rightarrow -\infty$
$L_{\text{IR}} = R^2/a \approx 1.3 \times 10^{26} \text{ m}$	T4	19–79 decades off, depending on which a is used
$\rho_{\text{DM}} = \frac{\hbar^2}{2m} \nabla\psi ^2$	T4	contains $\frac{1}{2}\rho v^2$; needs $m \sim 10^{-23} \text{ eV}$
Axiom I (non-amenable $\mathcal{B}$)	T4	every abelian group is amenable
$S_{\text{LF}} = 2\sqrt{2}$	T3	<i>arithmetically correct</i> ; but it is CHSH, not an LF inequality

Table 1: Claim ledger for the CPF manuscript. Tiers never promote.

Proof. Substituting (1) into (2), $E_{\text{self}} = \frac{e^2}{4} \cdot \frac{L}{4\pi\epsilon_0 R} = \frac{e^2 L}{16\pi\epsilon_0 R}$. Setting this to $m_e c^2$ and using (3) to write $1/R = 2m_e c/\hbar$ gives $\frac{e^2 L}{8\pi\epsilon_0 \hbar} = c$, i.e. $L \cdot \frac{e^2}{4\pi\epsilon_0 \hbar c} \cdot \frac{1}{2} = 1$, so $L\alpha = 2$. Verified symbolically in `cpf_audit.py`, check C1. $\square$

Remark 1 (This correction is already in the repository). *Proposition 1 is not new here. It is equation (alpha_ann) of papers/notes/Mobius_Ribbon_Capacitance.tex (2026-07-22), which states $\alpha_{\text{ann}}^{-1} = \frac{1}{2}(\ln(8R/a) + 1)$ and records that matching CODATA then forces $a/R = 8 \exp(1 - 2\alpha^{-1}) \approx 2 \times 10^{-118}$. The CPF manuscript reproduces the parent note’s uncorrected form and its 137.036, without citing either the correction or the conformal/BIE revisions that showed $\alpha^{-1} = \mathcal{O}(1)$ at moderate aspect ratio.*

Remark 2 (Provenance of the logarithm). $\ln(8R/a)$ is the Kelvin–Maxwell thin-ring formula’s log term. In the standard ring inductance $L = \mu_0 R [\ln(8R/a) - 7/4]$ the additive constant is $-7/4$ (or -2 for the external-inductance convention). The manuscript’s $+1$ is neither derived nor attributed. Since α^{-1} depends on that constant additively, the choice is a second free knob.

Remark 3 (Circularity). *Even granting $\alpha^{-1} = L$, the step*

$$\frac{a}{R} = 8e^{-136.035999171} \implies \alpha^{-1} = \ln(8/(a/R)) + 1 = 137.035999171$$

is an identity, not a derivation: $\ln(8/(8e^{-x})) + 1 = x + 1$ for any x . The target value is inserted in the premise and recovered in the conclusion. The asserted mechanism — “the eigenvalue gap of the G-field” with “GQRE renormalization” — is stated without a computation, and the Gravity-from-Entropy G-field is algebraically constrained rather than obeying an independent wave equation, so the discrete spectrum the argument needs does not follow from the GfE action without further input.

3 The cutoff a : four mutually exclusive values

The manuscript’s central claim is that it has *zero free parameters*, resting on the assertion (§3) that “the cutoff a is not free.” The manuscript places four separate requirements on a . They do not agree. Three points follow.

(i) **The manuscript contradicts itself within one section.** §3 asserts $\tau = ia/R = i/2$, hence $a/R = 1/2$, and then five lines later asserts $a/R = 8e^{-136.036} \approx 6.7 \times 10^{-59}$. These differ by 58 decades. On $a/R = 1/2$ the manuscript’s own formula gives $\alpha^{-1} = \ln 16 + 1 = 3.77$.

(ii) **The α leg and the L_{IR} leg are incompatible.** Taking the manuscript’s stated a , $L_{\text{IR}} = R^2/a = 2.9 \times 10^{45} \text{ m}$, 19 decades above the Hubble radius. Taking the a that the *corrected* α

Constraint (manuscript source)	a/R	α^{-1} (corrected)	$L_{\text{IR}} = R^2/a$ [m]
$\tau = ia/R = i/2$ (§3)	5.0×10^{-1}	1.886	3.9×10^{-13}
$a/R = 8e^{-136.036}$ (§3)	6.66×10^{-59}	68.518	2.9×10^{45}
CODATA match on Prop. 1	2.04×10^{-118}	137.035999177	9.5×10^{104}
$L_{\text{IR}} = 1.3 \times 10^{26}$ m (§8)	1.49×10^{-39}	46.242	1.3×10^{26}

Table 2: The four requirements the manuscript places on its single scale parameter, evaluated at $R = \hbar/(2m_e c) = 1.9308 \times 10^{-13}$ m. They span 117.4 decades. `cpf_audit.py`, check C2.

relation requires, $L_{\text{IR}} = 9.5 \times 10^{104}$ m — 79 decades off. Conversely, forcing $L_{\text{IR}} = 1.3 \times 10^{26}$ m gives $\alpha^{-1} = 46.24$. The two “derivations” cannot both be run.

(iii) **“UV complete” does not survive either.** The a demanded by the corrected α relation is 3.9×10^{-131} m, i.e. 10^{-96} Planck lengths. A UV cutoff 96 decades below ℓ_P is not a completion.

Proposition 2. *The Constraint Projection Framework has exactly one free parameter, a/R , fitted independently in §3 and §8. The parameter-count table of Appendix B is therefore incorrect as stated, and the comparison it draws with the Standard Model’s 19 is not like-for-like: the framework produces no lepton or quark mass, no mixing angle, and no coupling other than α .*

4 Axiom III has no model

Lemma 1 (The surface is forced). *A closed compact smooth non-orientable surface with $w_1 \neq 0$ whose orientation double cover is T^2 is the Klein bottle $K = T^2/\langle\tau\rangle$, $\tau(x, y) = (x + \frac{1}{2}, -y)$. Its $\pi_1 = \mathbb{Z} \rtimes \mathbb{Z}$ and $H_2(K; \mathbb{Z}) = 0$, so clauses (1)–(2) of Axiom III are satisfied and determine $\mathcal{M}$ uniquely.*

Proposition 3 (Clause (3) is unsatisfiable). *There is no $\phi \in \text{Diff}(K)$ with $\phi_* = (\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix})$.*

Proof. Every diffeomorphism of K lifts to T^2 and must normalise the deck group $\{1, \tau\}$. That group is $\mathbb{Z}/2$, so normalising means commuting. On $H_1(T^2) = \mathbb{Z}^2$ the linear part of τ is $D = \text{diag}(1, -1)$, so the image of $\text{MCG}(K)$ in $\text{GL}(2, \mathbb{Z})$ lies in the centraliser of D . Solving $[M, D] = 0$ forces M diagonal; the integer diagonal matrices of determinant ± 1 number four,

$$\{\text{diag}(1, 1), \text{diag}(-1, -1), \text{diag}(1, -1), \text{diag}(-1, 1)\} \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2,$$

in agreement with Lickorish’s computation $\text{MCG}(K) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2$. This group is *finite*. But $\phi_*^n = (\begin{smallmatrix} 1 & 2n \\ 0 & 1 \end{smallmatrix})$, so ϕ_* is parabolic of infinite order, and a finite group contains no element of infinite order. Independently, $\phi_* D \phi_*^{-1} = (\begin{smallmatrix} 1 & -4 \\ 0 & -1 \end{smallmatrix}) \neq D$, so ϕ_* does not descend to K at all. Verified as exact integer arithmetic in `cpf_audit.py`, check C3. $\square$

Remark 4 (The weaker reading fails too). *One might retreat to “some mapping class of $\mathcal{M}$ has $\text{Tr}(\phi_*) = 2$.” Within the centraliser the available traces are $\{+2, -2, 0, 0\}$, and $+2$ is attained only by the identity — which is not a Dehn twist, effects no half-twist, and supplies no self-linking number. The identification $\text{Tr}(\phi_*) = 2 = \text{Sl}$ has no carrier under either reading.*

Proposition 3 is the structural kill. Everything the manuscript derives from $\text{Tr}(\phi_*) = 2$ — $\tau = i/2$ in §3, g and s in §4, and the “Dehn twist condition” $\lambda_{\text{UV}} \lambda_{\text{IR}} = 1/R^4$ in §8 — rests on an axiom that no surface satisfies.

Remark 5 (A second, independent obstruction in §3). *§3 computes an electrostatic capacitance for $\mathcal{M}$. The Klein bottle admits no embedding in $\mathbb{R}^3$ — only immersions with self-intersection — so there is no conductor to charge. It is also closed, so it has no boundary and no “double-cover annulus.” The formula actually used is the thin-ring one, which describes a different object.*

5 §4: spin and g restate two logged kills

The manuscript asserts $g = Sl = 2$ and $s = Sl/4 = 1/2$. Both were tested and falsified in this repository on 2026-07-26; see `docs/Elimination_Ledger.md`.

(a) **$g = Sl$ was killed by the parity law.** For torus curves $\sigma = (-1)^{p+q} = (-1)^{Sl+1}$, measured over a control family of framed circles. The $Sl = 0$ round circle is spinorial in exactly the same sense as the $|Sl| = 2$ screw. The spin-relevant content of Sl is *one parity bit*, and 2 and 0 are the same bit; the value 2 does no work. The mechanism of the original error — matching the integer 2 across two formalisms in which it arises for unrelated reasons (pq on one side, $|\pi_1(\text{SO}(3))|$ on the other) — is reproduced verbatim in the CPF manuscript, now with a third unrelated 2 (the trace of a parabolic) matched to the same integer.

(b) **The magnitude route is closed by a no-go.** The successor test computed μ against $\langle L \rangle$ directly: both are proportional to the same vector area $A = \frac{1}{2} \oint \mathbf{r} \times d\boldsymbol{\ell}$, so $\mu/\langle L \rangle = q/2m$ and $g = 1$ *exactly, for every closed curve* — whatever the winding, framing, twist or throat. No amount of topology moves g without decoupling where the charge sits from where the mass sits, which the manuscript does not do.

(c) **Two further errors.** The repository's own computation of the $(2,1)$ centerline gives $\text{Tw} + \text{Wr} = -2.000000$; on $s = Sl/4$ that is $s = -1/2$. And a (p, q) torus knot with $q = 1$ is the *unknot*; the parent note already carries this correction. The factor 4 in $s = Sl/4$ is not derived anywhere: Finkelstein–Rubinstein yields a $\mathbb{Z}/2$ statistics sector, not a magnitude — which is precisely the gap the ledger recorded when it killed (a).

6 §7: the curvature functional cannot see an off-line zero

The manuscript's functional is

$$k(\varepsilon) = 2\varepsilon \sum_{\rho} \frac{1}{(\frac{1}{2} - \beta)^2 - \varepsilon^2},$$

with the claim that $k = 0$ requires $\beta = \frac{1}{2}$ for all ρ , hence RH, hence $\Omega_k = 0$. Three independent failures:

(i) **Blindness.** β enters only through $(\frac{1}{2} - \beta)^2$, which is exactly the invariant of the functional equation's involution $\beta \mapsto 1 - \beta$. The zero set of ζ is symmetric under that involution, so every off-line zero arrives with a mirror partner contributing identically. $k(\varepsilon)$ *takes the same value on a critical-line configuration and on its reflection*, and cannot distinguish RH from its negation. The instrument confirms this numerically: at $\varepsilon = 0.1$ over 50 zeros, $\beta = 0.5 \pm 0.2$ both give $k = +333.33$.

(ii) **Inverted sign.** On the critical line each summand equals $-1/\varepsilon^2$ — not zero. For N on-line zeros $k(\varepsilon) = -2N/\varepsilon$, which diverges as $N \rightarrow \infty$ and vanishes for no finite ε . RH makes k maximally divergent. The stated criterion is not merely unproved; it points the wrong way.

(iii) **Empirical direction.** Even a repaired functional would not make $\Omega_k = 0 \Leftrightarrow \text{RH}$ testable. Ω_k is measured as 0.0007 ± 0.0019 ; no measurement establishes an exact zero. A biconditional here makes a theorem of arithmetic contingent on CMB data, and would let a future curvature detection refute RH.

Remark 6. *This repository nowhere claims the Riemann Hypothesis proved, and constructs no Hilbert–Pólya operator; every `hp_knife` script says so in its own output. §7 claims an equivalence, which is a proof claim in both directions. The nearest defensible statement in the corpus is Paper B's: $\text{RH} \Rightarrow \text{stationarity}$ is proved by AM–GM, and the converse is conditional on an unproved minimum-gap bound verified only to $N = 200$.*

7 §5: the arithmetic is right; the identification is not

This is the one section whose displayed number survives its own computation. With $A(\theta) = \cos \theta \sigma_z + \sin \theta \sigma_x$ on $|\Phi^+\rangle$ and the stated angles, the instrument returns $E(A_2, B_2) = E(A_2, B_3) =$

$$E(A_3, B_2) = +1/\sqrt{2}, \quad E(A_3, B_3) = -1/\sqrt{2}, \quad \text{and}$$

$$S = 2\sqrt{2} = 2.8284271247\dots,$$

saturating the Tsirelson bound to machine precision. **Confirmed, T3.**

The identification does not survive. Only settings $\{2, 3\}$ appear on either wing. The setting that makes an Extended Wigner’s Friend Scenario what it is — $x = 1$ / $y = 1$, “open the lab and read the friend’s already-recorded outcome” — enters no term of the sum. With two settings per wing this expression *is* CHSH; its local bound is 2 because that is the CHSH bound, not because of anything about observers. Consequently:

- §5 demonstrates ordinary Bell nonlocality, experimentally established since 1982, not a Local Friendliness violation. The LF facets of Bong *et al.* involve the $x = 1$ row precisely because that is where Absoluteness of Observed Events enters the derivation.
- “Falsifying AOE” overstates even a genuine LF violation. The LF theorem falsifies the *conjunction* of Absoluteness of Observed Events, Locality and No-Superdeterminism. No single conjunct is singled out, and this is stated explicitly in the source the manuscript cites.
- The C_6/D_6 axial frame, the projection of $|\Phi^+\rangle$ onto it, and the super-observer structure are inert: none appears anywhere in the algebra that produces $2\sqrt{2}$.

8 §6: the dark-matter term double counts, and needs an exotic scalar

With $\psi = Re^{iS/\hbar}$, $\nabla\psi = (R' + iRS'/\hbar)e^{iS/\hbar}$ and

$$\frac{\hbar^2}{2m}|\nabla\psi|^2 = \underbrace{\frac{\hbar^2 R'^2}{2m}}_{\text{amplitude gradient}} + \underbrace{\frac{R^2 S'^2}{2m}}_{= \frac{1}{2}\rho v^2}, \quad \rho = R^2, \quad v = S'/m,$$

an exact split (`cpf_audit.py`, check C7).

(i) Double counting. The second term is the ordinary kinetic energy density of the visible matter itself. The manuscript’s $T_{00} = \rho_{\text{vis}}c^2 + \rho_{\text{DM}}c^2$ then counts it a second time, as dark.

(ii) The $\Re/\Im$ story is wrong. $|\nabla\psi|^2$ is not a function of $\Im\psi$ alone, and the $\Re/\Im$ split is not gauge invariant — a global $U(1)$ phase rotates one into the other. Electromagnetic coupling enters through the covariant derivative, not through “the real component.”

(iii) The scale. Quantum pressure shapes a rotation curve only when the de Broglie wavelength is galactic. At $v = 200$ km/s and $L = 1$ kpc the required mass is 1.7×10^{-59} kg $= 9.6 \times 10^{-24}$ eV/ c^2 — the fuzzy-dark-matter window, i.e. an ultralight scalar, which is exactly the “exotic particle” the manuscript claims to avoid. For the electron mass, $\lambda_{\text{dB}} = 5.8 \times 10^{-10}$ m, some 29 decades too short-ranged to matter at kpc scales. So the claim fails either way: at any Standard Model mass the effect is absent; at the mass that works, the particle is exotic.

No Jeans-equation solution, rotation curve, or galaxy dataset appears in §6; “matching observations” is asserted, not shown.

9 Axioms I–II and the bibliography

Axiom I is false under every reading. $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^\times$ is malformed: $\mathbb{Q}^\times$ is the multiplicative group and does not act on the additive adeles by translation. The two standard objects are $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ (compact, additive) and the idele class group $\mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$. Both are *abelian*, hence amenable — every abelian group is amenable, by Markov–Kakutani. The first is moreover compact, so it carries a translation-invariant Haar *probability* measure, which is countably additive and therefore finitely additive. The axiom asserts that no such measure exists. Non-amenability is load-bearing for the manuscript’s “non-amenable information reservoir” framing, and it is unavailable here.

Axiom II is ill-posed. The kernel maps onto $\mathbb{R}^{3,1}$, but $\delta^{(3)}(\mathbf{r}(u, v) - \mathbf{x})$ fixes only three coordinates; nothing in the kernel produces the time direction. A two-parameter integral against a three-dimensional delta is also generically distributional rather than a function, so “Fredholm” is not established.

Bibliography.

- [Bianconi2024] arXiv:2401.12345 is a placeholder identifier. The paper is G. Bianconi, “Gravity from entropy,” arXiv:2408.14391, Phys. Rev. D **111**, 066001 (2025).
- [CODATA2022] “E. Tiesinga et al., Rev. Mod. Phys. **94**, 035002 (2023)” matches no CODATA article of record. CODATA 2018 is Tiesinga, Mohr, Newell & Taylor, Rev. Mod. Phys. **93**, 025010 (2021); CODATA 2022 is Mohr, Newell, Taylor & Tiesinga, Rev. Mod. Phys. **97**, 025002 (2025), with $\alpha^{-1} = 137.035999177(21)$.
- [Proietti2019] and [Finkelstein1968] check out as cited.

10 What survives, and what repair would take

Survives.

1. $S = 2\sqrt{2}$ under the stated operators and state (**T3**, confirmed). It is CHSH at Tsirelson saturation — a correct computation of a known quantity.
2. The identification $\mathcal{M} =$ Klein bottle from clauses (1)–(2) of Axiom III is correct and, as the manuscript says, unique. It is the one piece of topology in the paper that does what it claims. Clause (3) is what fails.
3. The observation that $\ln(8R/a)$ is the right functional form for a thin-ring self-energy is inherited correctly from the parent note; only the prefactor, the additive constant, and the cutoff are wrong.

What repair would take, leg by leg.

- α : adopt Proposition 1, then *derive* a/R from something that is not α — and accept whatever number comes out. The repository’s conformal and BIE revisions give $\mathcal{O}(1)$ at moderate aspect, so the honest current state is that this route does not reach 137.
- *Axiom III*: either drop clause (3), or replace $\mathcal{M}$ with a surface whose mapping class group actually contains an infinite-order parabolic (genus ≥ 1 orientable, or a non-orientable surface of higher genus) — but then the T^2 double cover, and with it the whole modular-parameter argument, is gone.
- g : the ledger’s standing successor question applies unchanged — **decouple where the charge sits from where the mass sits**. That is the only known route past $g = 1$, and no framing, twist or trace matching will substitute for it.
- §7: a functional that detects off-line zeros must be *odd* under $\beta \mapsto 1 - \beta$, not even. Any construction whose β -dependence factors through $(\frac{1}{2} - \beta)^2$ is blind for the same reason, so this is a structural constraint on the whole approach, not a fixable sign.
- §5: include the $x = 1$ / $y = 1$ rows and test an actual LF facet. The result would then be about observers rather than about CHSH — and would falsify a conjunction, which must be stated as such.
- §6: if the ultralight-scalar mass is accepted, the model becomes fuzzy dark matter, which is a real and testable proposal — but it is then a proposal *with* an exotic particle, and inherits that literature’s constraints.

On the framing. The manuscript’s closing line, “the ledger is sealed,” inverts this program’s discipline. The Elimination Ledger is append-only precisely so that it is never sealed; its entries are kills aimed at the program’s own load-bearing claims, kept because they bound the space. Two of the claims this manuscript advances as derivations are entries in that ledger. A framework that declares zero free parameters while fitting one parameter four ways, and declares completeness while restating falsified results, is describing the failure mode this corpus names as primary: *overclaiming*.

Verification

Instrument: `code/constraint_projection/cpf_audit.py` (checks C1–C8; `numpy`, `sympy`, `mpmath`).
Artifact: `docs/constraint_projection_audit.json`. Ledger entry: `docs/Elimination_Ledger.md`, 2026-08-29. Constants: CODATA 2022, $\alpha^{-1} = 137.035999177(21)$. Prior repository results relied on: `papers/notes/Mobius_Ribbon_Capacitance.tex` (the corrected annulus relation), `papers/notes/Framing_Trans` and `code/framed_unknot/` (the parity law and the $g = 1$ no-go).

References

- [1] P. J. Mohr, D. B. Newell, B. N. Taylor and E. Tiesinga, “CODATA recommended values of the fundamental physical constants: 2022,” *Rev. Mod. Phys.* **97**, 025002 (2025); [arXiv:2409.03787](#).
- [2] G. Bianconi, “Gravity from entropy,” *Phys. Rev. D* **111**, 066001 (2025); [arXiv:2408.14391](#).
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- [4] M. Proietti *et al.*, “Experimental test of local observer independence,” *Sci. Adv.* **5**, eaaw9832 (2019).
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- [8] W. Hu, R. Barkana and A. Gruzinov, “Fuzzy cold dark matter: the wave properties of ultralight particles,” *Phys. Rev. Lett.* **85**, 1158 (2000).